

Biology Quiz — Cell Structure & Function

Name: _____ Date: _____ Score: _____ / 100

Section A Multiple Choice (40 pts)

1. Which organelle is responsible for producing ATP through cellular respiration?

- ☐ A. Nucleus
- ☐ B. Mitochondria
- ☐ C. Ribosome
- ☐ D. Golgi apparatus

2. The cell membrane is primarily composed of:

- ☐ A. Proteins only
- ☐ B. Carbohydrates only
- ☐ C. A phospholipid bilayer with embedded proteins
- ☐ D. Cellulose

3. Which of the following is found in plant cells but NOT in animal cells?

- ☐ A. Mitochondria
- ☐ B. Cell membrane
- ☐ C. Cell wall
- ☐ D. Nucleus

4. Ribosomes are responsible for:

- ☐ A. DNA replication
- ☐ B. Protein synthesis
- ☐ C. Lipid production
- ☐ D. Energy storage

5. The process by which water moves across a semi-permeable membrane is called:

- ☐ A. Active transport
- ☐ B. Diffusion
- ☐ C. Osmosis
- ☐ D. Endocytosis

Section B Short Answer (40 pts)

6. (10 pts) Describe the structure and function of the cell nucleus. Include the role of the nuclear envelope and nucleolus in your answer.

7. (10 pts) Explain the difference between prokaryotic and eukaryotic cells. Give one example of each.

8. (10 pts) What is the function of the endoplasmic reticulum (ER)? Distinguish between rough ER and smooth ER.

9. (10 pts) Describe how mitochondria produce energy. Why are they called the 'powerhouse of the cell'?

Section C Diagram Label (20 pts)

10. (20 pts) Label the parts of the animal cell diagram below. Write the name of each organelle on the blank lines

